

Rossby Wave Ray-tracing: A Modern Approach

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Abstract

1 Introduction

The 1993 paper titled *Rossby Wave Propagation on a Realistic Longitudinally Varying Flow* by Hoskins and Ambrizzi [1] suggests a model for Rossby waves are large-scale meanders in the jet stream that arise from Earth's rotation. They are responsible for linking weather and climate across distant regions a phenomenon known as teleconnection. For example, El Nio warming in the tropical Pacific can influence winter temperatures in North America and Europe via Rossby wave trains.

Hoskins and Ambrizzi (1993) showed that these waves do not propagate randomly. Instead, the background wind structure creates "waveguides" preferred paths that channel wave energy. Their key tool was geometric ray tracing, adapted from optics.

2 Theory

2.1 The Stationary Wavenumber K_s

The central concept in the paper is the stationary wavenumber K_s . It tells us, at each location in the atmosphere, what wavelength a Rossby wave must have to remain stationary relative to the ground. The formula is:

$$K_s^2 = \frac{\beta_M - \partial^2 U / \partial y^2}{U} \quad (1)$$

Where U is the background zonal wind and β_M is the gradient of Earth's vorticity in Mercator coordinates.

When K_s is large and uniform over a region, that region acts as a waveguide: waves of the appropriate scale are trapped and can travel long distances. When K_s^2 becomes negative, waves cannot propagate at all (they are "evanescent").

2.2 Ray Tracing

Ray tracing treats a wave packet as a particle whose position (x, y) and wavenumber (k, l) evolve according to Hamilton's equations. The key idea is that waves bend toward regions of higher K_s , similar to how light bends in a medium with varying refractive index. By integrating these equations forward from a source location, we can predict where wave energy will go.

3 Data and Methods

3.1 Data

- Source data:
 - ERA5 reanalysis
 - Spatial variables: zonal (u) and meridional (v) winds at 300 hPa
 - Time variable: Monthly from January 1940 to April 2026
 - Spatial resolution: $0.25^\circ \times 0.25^\circ$
- Used data:
 - Time-averaged over December-February (referred to as DJF)
 - Spatial resolution: $2.5^\circ \times 2.5^\circ$

3.2 Computing K_s

3.3 Ray Tracing Implementation

4 Results

4.1 Comparison with Hoskins & Ambrizzi (1993)

5 Discussion

References

- [1] Brian J. Hoskins and Tercio Ambrizzi. Rossby wave propagation on a realistic longitudinally varying flow. *Journal of Atmospheric Sciences*, 50(12):1661 – 1671, 1993.